

SKILL2 K. GAYATHRI SRIVALLI

The image displays two screenshots of the MySQL Workbench interface, showing SQL queries and their execution results.

Top Screenshot:

- Query 1:**

```
1 CREATE DATABASE bookflow_dh;
2 USE bookflow_dh;
3 CREATE TABLE Books
4 ( book_id INT PRIMARY KEY,
5  title VARCHAR(100) NOT NULL,
6  isbn VARCHAR(20) UNIQUE,
7  published_year INT CHECK (published_year < 2027)
8 );
9
10 INSERT INTO Books (book_id, title, isbn, published_year)
11 VALUES (1, 'The Great Gatsby', '9780743273565', 1925),
12 (2, 'To Kill a Mockingbird', '9780061120084', 1960),
13 (3, '1984', '9780451524935', 1949);
14 SELECT * FROM Books;
```
- Output:**

#	Time	Action	Message	Duration / Fetch
39	21:14:02	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(20) UNIQUE, published_year INT CHECK (published_year < 2027))	0 row(s) affected	0.062 sec
40	21:14:11	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (1, 'The Great Gatsby', '9780743273565', 1925), (2, 'To Kill a Mockingbird', '9780061120084', 1960), (3, '1984', '9780451524935', 1949);	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
41	21:14:11	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
42	21:14:53	INSERT INTO Members (member_id, full_name, email) VALUES (101, 'John Smith', 'john.smith@email.com'), (102, 'Emma Wilson', 'emma.wilson@email.com'), (103, 'Michael Brown', 'michael.brown@email.com');	Error Code: 1146. Table 'bookflow_dh.members' doesn't exist	0.000 sec
43	21:15:00	CREATE TABLE Members (member_id INT PRIMARY KEY, full_name VARCHAR(100), email VARCHAR(100) UNIQUE)	0 row(s) affected	0.078 sec
44	21:15:07	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE)	0 row(s) affected	0.110 sec
45	21:15:12	INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES (1, 101, 1, '2023-10-27'), (2, 102, 2, '2023-10-28'), (3, 103, 3, '2023-10-29');	Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails (bookflow_dh.loans, CONSTRAINT fk_loan_member FOREIGN KEY (member_id) REFERENCES members (member_id))	0.016 sec
46	21:15:16	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l JOIN members m ON l.member_id = m.member_id JOIN Books b ON l.book_id = b.book_id	0 row(s) returned	0.000 sec / 0.000 sec

Bottom Screenshot:

- Query 1:**

```
13 (3, '1984', '9780451524935', 1949);
14 SELECT * FROM Books;
15
16 CREATE TABLE Members
17 ( member_id INT PRIMARY KEY,
18  full_name VARCHAR(100),
19  email VARCHAR(100) UNIQUE
20 );
21 INSERT INTO Members (member_id, full_name, email) VALUES
22 (101, 'John Smith', 'john.smith@email.com'),
23 (102, 'Emma Wilson', 'emma.wilson@email.com'),
24 (103, 'Michael Brown', 'michael.brown@email.com');
25 SELECT * FROM Members;
26
```
- Output:**

#	Time	Action	Message	Duration / Fetch
47	21:15:24	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY published_year	3 row(s) returned	0.000 sec / 0.000 sec
48	21:15:36	CREATE TABLE Donation_History (donation_id INT PRIMARY KEY, book_id INT, donor_name VARCHAR(100), donation_date DATE)	0 row(s) affected	0.078 sec
49	21:15:36	START TRANSACTION	0 row(s) affected	0.000 sec
50	21:15:36	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (4, 'Animal Farm', '9780451524935', 1949);	1 row(s) affected	0.000 sec
51	21:15:36	INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date) VALUES (1, 4, 'John Smith', '2023-10-27'), (2, 4, 'Emma Wilson', '2023-10-28'), (3, 4, 'Michael Brown', '2023-10-29');	1 row(s) affected	0.016 sec
52	21:15:36	COMMIT	0 row(s) affected	0.000 sec
53	21:15:42	CREATE INDEX idx_books_isbn ON Books(isbn)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.109 sec
54	21:15:42	SELECT * FROM Books WHERE isbn = '9780451524935' LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- bookdonation_db
- bookflow_db
- bookflow_db
- capacity_connect
- capacity_connect_db
- data_d
- digitallibrary
- sakila
- students_db
- sys
- world

Query 1 SQL File 1* SQL File 5* SQL File 6* SQL File 7* SQL File 10* SQL File 11* x

Limit to 1000 rows

```

26
27 CREATE TABLE Loans (
28   loan_id INT PRIMARY KEY,
29   member_id INT, book_id INT,
30   loan_date DATE,
31   FOREIGN KEY (member_id) REFERENCES Members(member_id),
32   FOREIGN KEY (book_id) REFERENCES Books(book_id)
33 );
34
35 INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
36 (1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),
37 (3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),
38 (5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),
39 (7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07');

```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Administration Schemas

Information

No object selected

Object Info Session

Output

#	Time	Action	Message	Duration / Fetch
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49	21:15:36	START TRANSACTION	0 row(s) affected	0.000 sec
50	21:15:36	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (4, 'Animal Farm', 9780451524935, 1945);	1 row(s) affected	0.000 sec
51	21:15:36	INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date) VALUES (1, 4, 'John Doe', '2025-01-01');	1 row(s) affected	0.016 sec
52	21:15:36	COMMIT	0 row(s) affected	0.000 sec
53	21:15:42	CREATE INDEX idx_books_isbn ON Books(isbn)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.109 sec
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Query 1 SQL File 1* SQL File 5* SQL File 6* SQL File 7* SQL File 10* SQL File 11* x

Limit to 1000 rows

```

39 (7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07');
40 (9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
41 SELECT * FROM Loans;
42
43 SELECT m.full_name AS Member_Name, b.title AS Book_Title
44 FROM Loans l
45 INNER JOIN Members m ON l.member_id = m.member_id
46 INNER JOIN Books b ON l.book_id = b.book_id;
47
48 SELECT published_year, COUNT(book_id) AS Total_Books
49 FROM Books
50 GROUP BY published_year
51 ORDER BY published_year;
52

```

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Query 1 SQL File 1* SQL File 5* SQL File 6* SQL File 7* SQL File 10* SQL File 11* x

Limit to 1000 rows

```

51 ORDER BY published_year;
52
53 CREATE TABLE Donation_History (
54     donation_id INT PRIMARY KEY,
55     book_id INT,
56     donor_name VARCHAR(100),
57     donation_date DATE,
58     FOREIGN KEY (book_id) REFERENCES Books(book_id) );
59 START TRANSACTION;
60 INSERT INTO Books (book_id, title, isbn, published_year)
61 VALUES (4, 'Animal Farm', '9780451526342', 1945);
62 INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)
63 VALUES (1, 4, 'Raj Kumar', CURDATE());
64 COMMIT;

```

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```

55     book_id INT,
56     donor_name VARCHAR(100),
57     donation_date DATE,
58     FOREIGN KEY (book_id) REFERENCES Books(book_id) );
59 START TRANSACTION;
60 INSERT INTO Books (book_id, title, isbn, published_year)
61 VALUES (4, 'Animal Farm', '9780451526342', 1945);
62 INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)
63 VALUES (1, 4, 'Raj Kumar', CURDATE());
64 COMMIT;
65
66 CREATE INDEX idx_books_isbn ON Books(isbn);
67 SELECT * FROM Books WHERE isbn = '9780451524935';

```

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